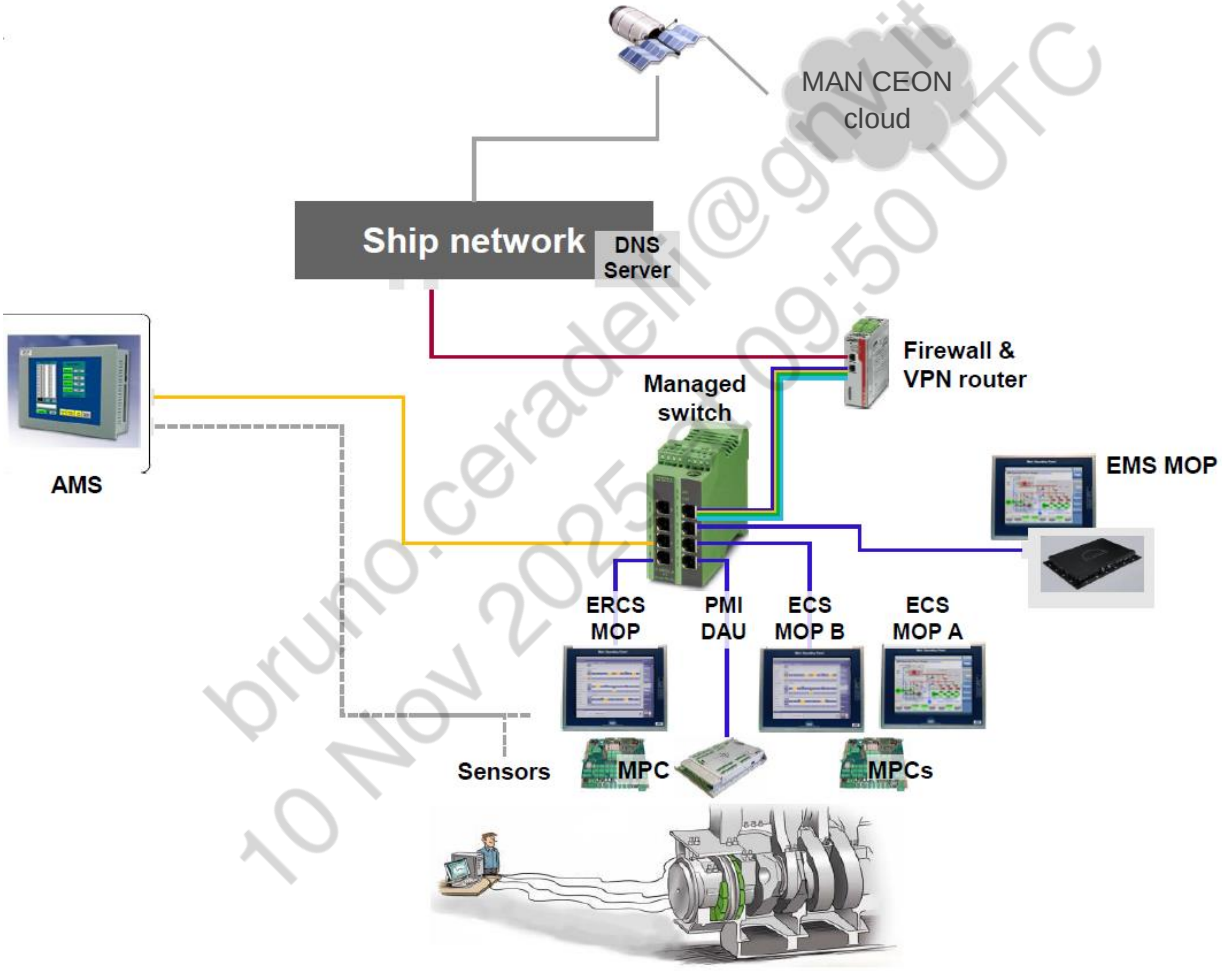
PRIMESERV ASSIST (PSA) INSTALLATION GUIDE

PrimeServ Digital Denmark



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10 Nov 2025 09:50 UTC

MAN Energy Solutions

v1.5 - June 2023

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Requirements

The following guide is only applicable for:

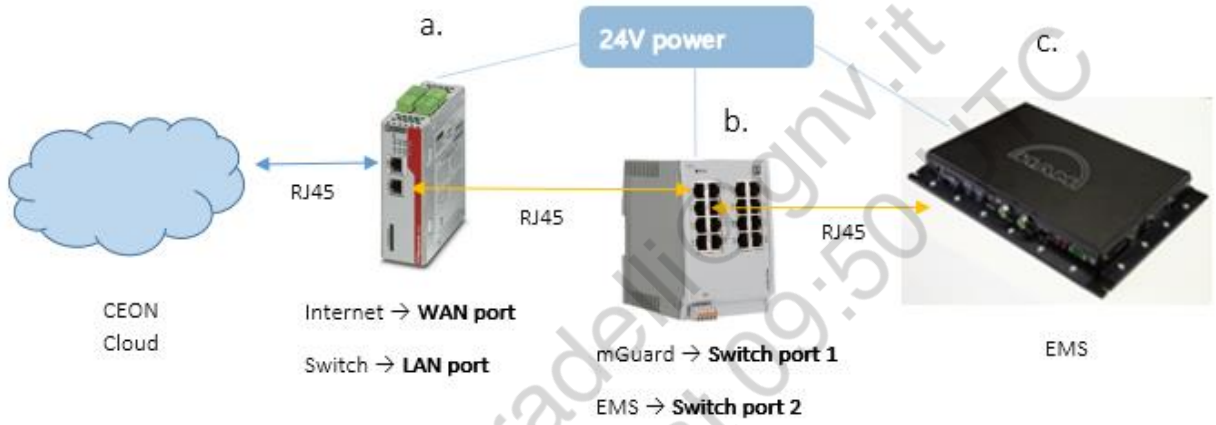
- MAN B&W ME-C engines with EC-MOP ECS version 1304-2.7 or higher.
- ME-B engines with EC-MOP ECS version 1403-1.12 or higher.
- All dual fuel engines.
- PMI-DAU 11 or higher.

If the requirements above are not met, additional hardware and attendance by MAN ES may be necessary to complete the installation.

Overview

The scope of this guide is to install or replace these 3 hardware units:		
a. **mGuard**	b. 16-port **MAN Managed switch**	c. Engine Monitoring System **EMS**

LAN connections:



Delivered parts to be checked
Standard hardware for single-engine installation: - 1 x mGuard (a) and Managed Switch (b) package - 1 x EMS-PC (c) - 1 x LAN Cable, RJ45, Shielded, (10m) - 1 x Multicore (Power cable) (5m) - 1 x Cable ties 4,6 X 360mm (100pcs) - 1 x Cable ties 2,4 X 180mm (100pcs) - 1 x Extra DIN-rail

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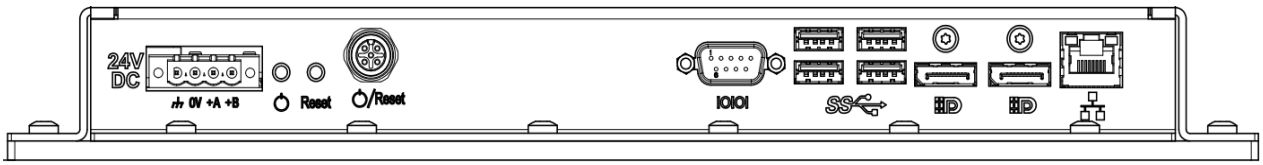
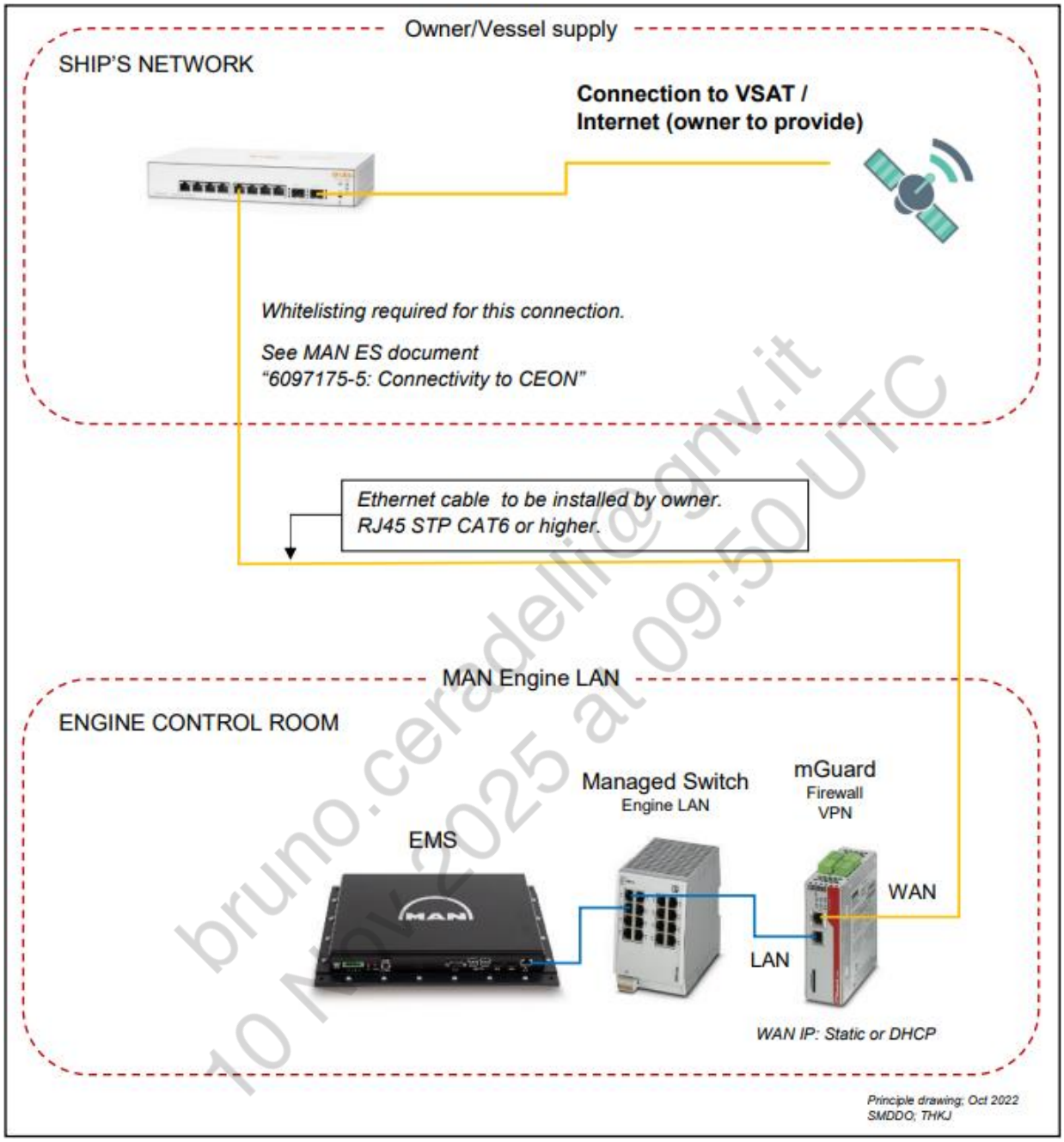
EMS connector layout:



PrimeServ Assist network overview



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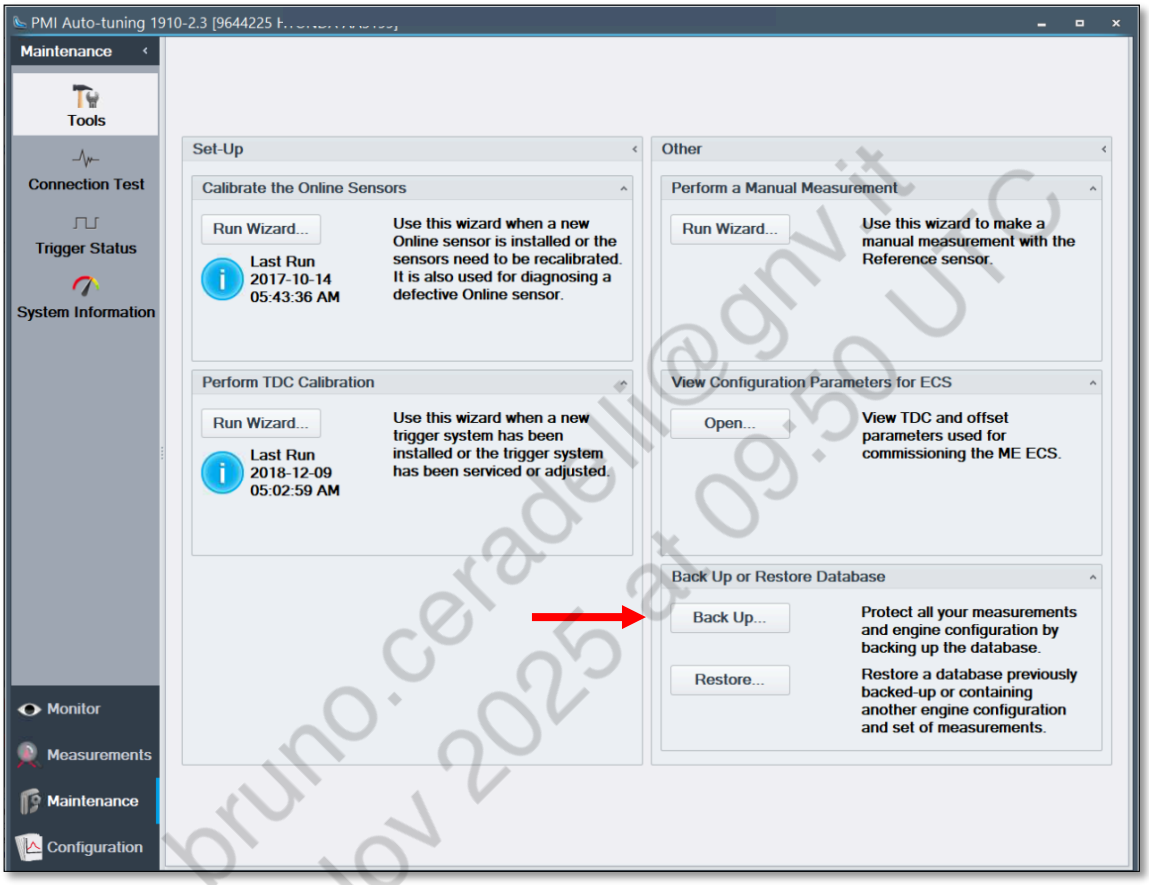
Installation steps 1-6:

1- PMI backup

Create a database backup (red arrow) from the PMI application and save it on a USB stick before turning off the PMI computer. This file will be used to set up the new EMS computer later (Step 4).



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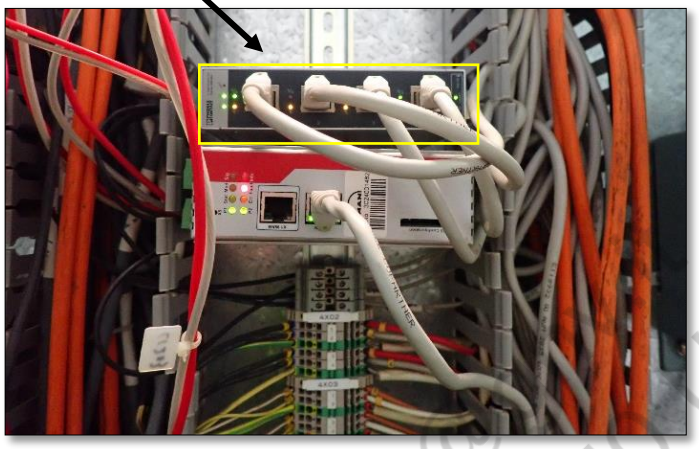
Important: the following steps must be done while the main engine is OFF.

2- MAN managed switch & mGuard

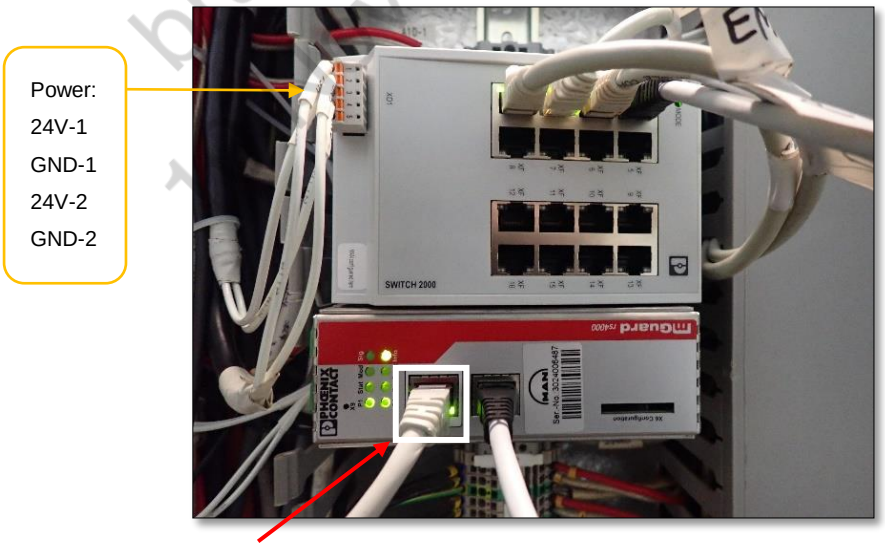
Locate the current managed switch and replace it with the new one (typically 4, 5 or 8 ports).



Mount the new 16-port MAN managed switch and new mGuard. Connect 24 VDC power.

Note: each port on the MAN managed switch is reserved to individual hardware so make sure to use the correct port.

The mGuard **LAN** port connects to **port 1** on the MAN managed switch.
Port 2: EMS computer
port 3: MOP B
port 4: PMI-DAU
port 5: ERCS (Tier III engines only)
port 6: AMS



Connect the mGuard WAN port to a VSAT switch with **an active internet connection.**
Consult your own IT for more information if necessary.

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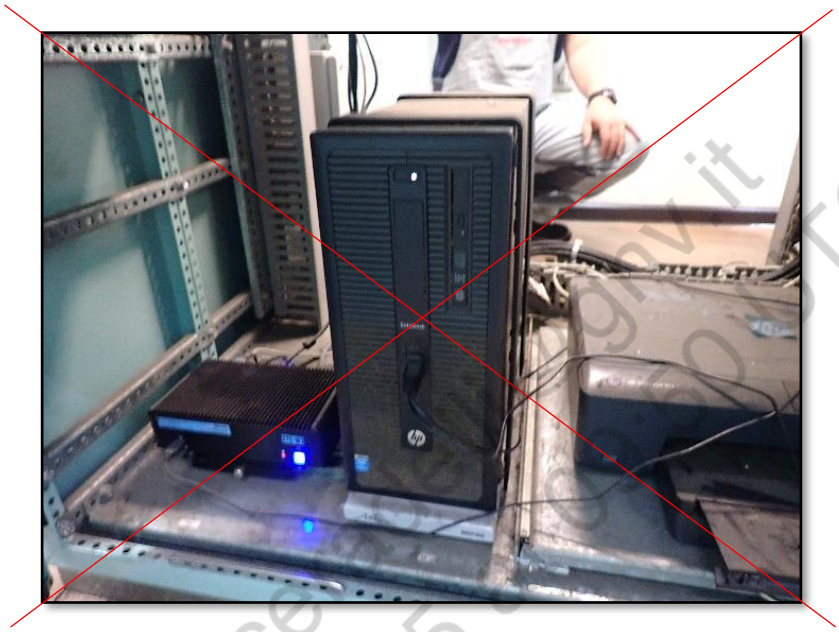
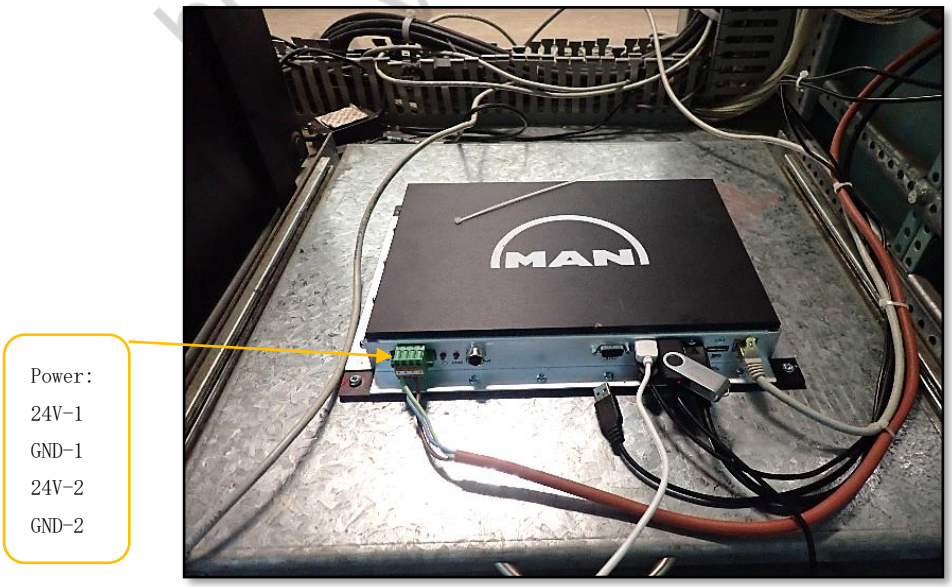
3- EMS installation

Turn off and remove the PMI+CoCos PC but leave the computer display in place – it will be used later with the EMS computer.

Note: we recommend taking a backup of the CoCos-EDS data for future reference.
Folder location: C:\users\public\MAN\CoCoS-EDS ME\[hull number]\LINEREC



Mount the new EMS in a suitable location. Re-connect all relevant cables, display, and establish 24 VDC power supply.
Connect the EMS' Ethernet cable to **port 2** of the MAN managed switch.



EMS installation continued ...

Feel free to choose a location for the EMS suitable for you.
Example of completed installations:

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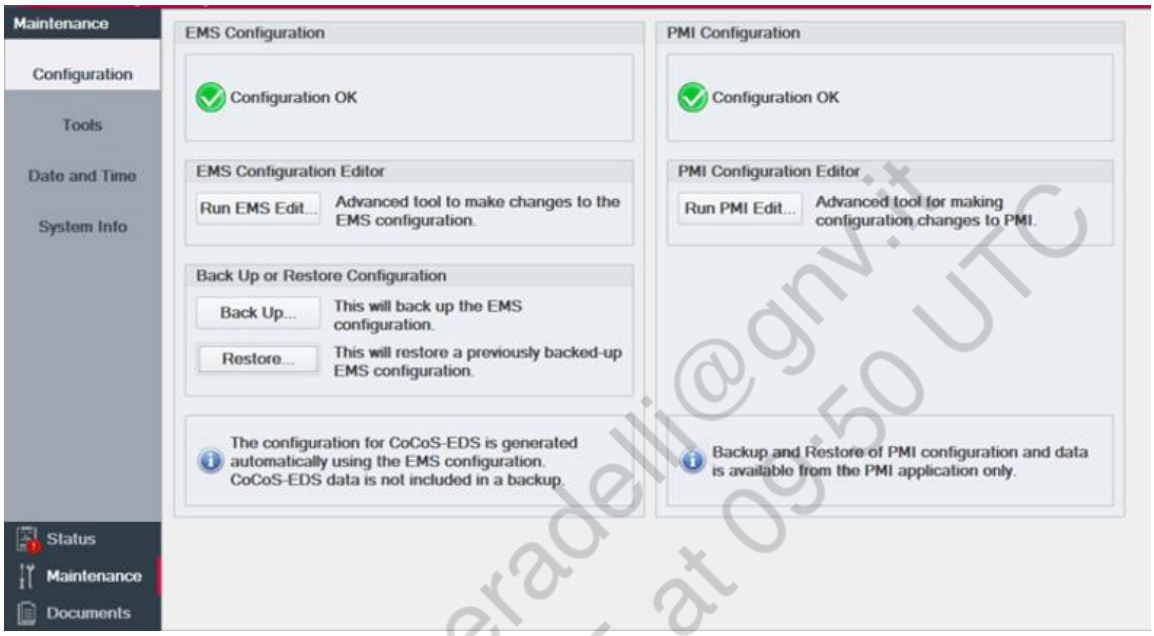
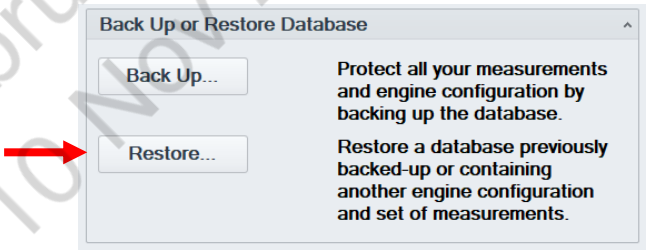
4- Starting up EMS for the first time

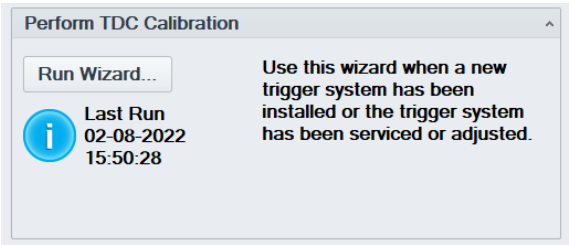
Once powered up, the EMS computer will automatically open the EMS application , which comes pre-installed on your EMS.
Open the EMS application window and go to *Maintenance | Configuration* … and check if both the EMS Configuration and PMI Configuration are okay, as shown below.



In case the PMI Configuration is missing or if you want to keep your historic PMI measurements, please Restore the backup file you saved on a USB stick (Step 1). The Restore function is located in the Maintenance menu in the PMI Auto-tuning application . After Restoring the program will re-start.



Note: If you choose to use the pre-installed PMI configuration on the new EMS, a new TDC calibration is necessary for the PMI Auto-tuning application to work. The calibration must be carried out while the main engine is running – please follow the instructions in the calibration wizard.



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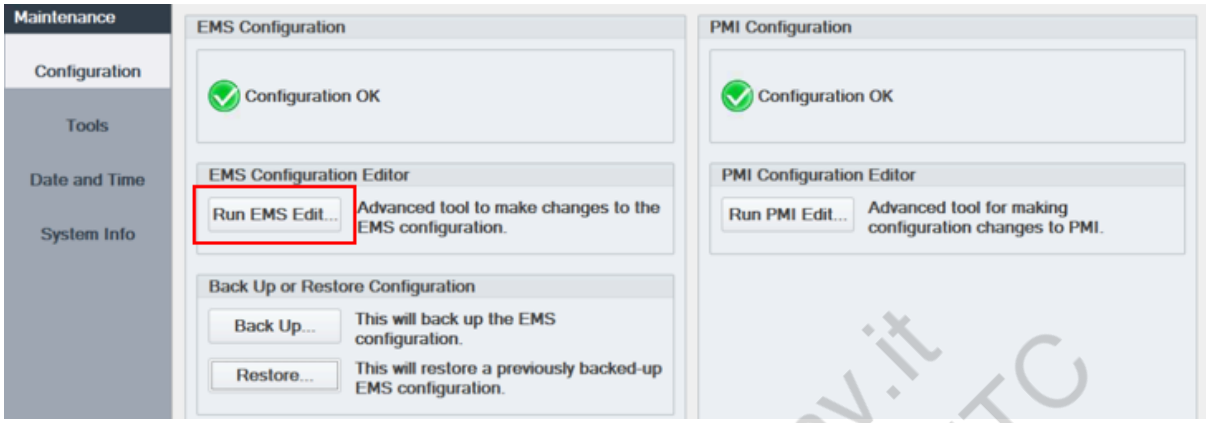
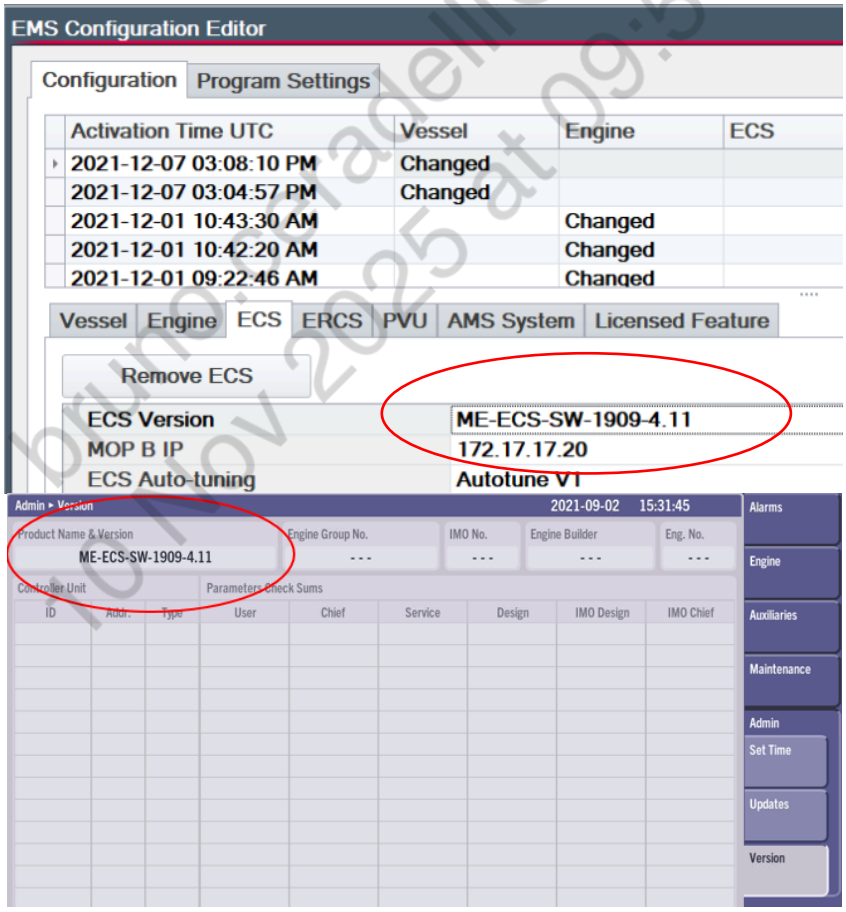
5- Check ECS version

Run **EMS Edit** in the EMS application .



Choose the **ECS** tab and compare the ECS version with the one installed on MOP-A and MOP-B. Choose the correct version from the drop-down menu so that they match.



Pres **OK** and close the EMS Configuration Editor.

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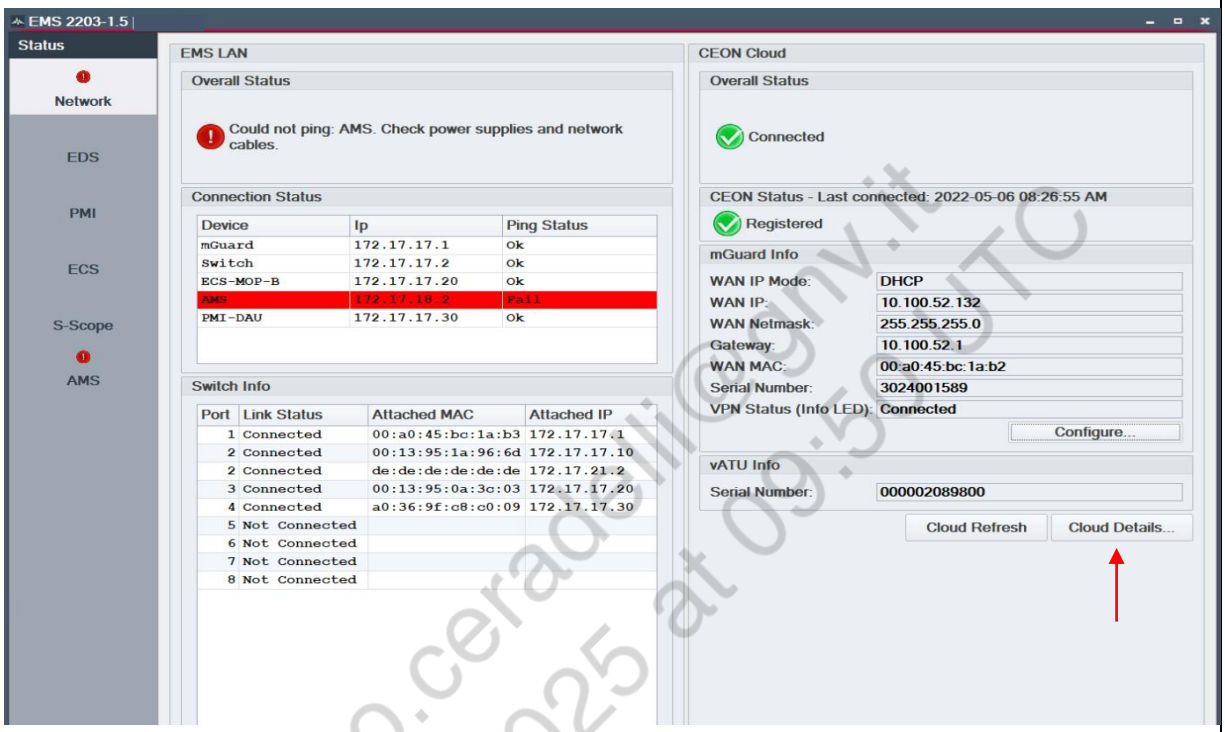
6- EMS connectivity

*See the **Network requirements for MAN CEON** on page 13 for more details.*

Go to the **Status | Network** page.

If the internet connection has been properly configured, your screen should look like this:



*Note: The AMS connection highlighted with red will only change to **OK** when the AMS maker has EMS-enabled their interface.*

Check that the mGuard network settings are correct and whether WAN IP Mode should be DHCP or Static (informed by vessel's IT). See how to change to static IP on the next page.

🟢 If CEON Cloud Overall Status is **Connected** & **Registered**, everything is fine and no further action is required.

Troubleshooting – CEON Cloud Overall Status:
🟡 *"various error messages" ...,*
Solution: check with your IT and VSAT provider that the relevant domain(s) has been whitelisted - see table on page 13.

🟡 If any other errors appear, press **Cloud Details** (red arrow), copy the text to Notepad and send it as a text-file along with a screenshot of the entire Network Status screen to the PrimeServ Remote Operation Center: 2stroke-connectivity@man-es.com

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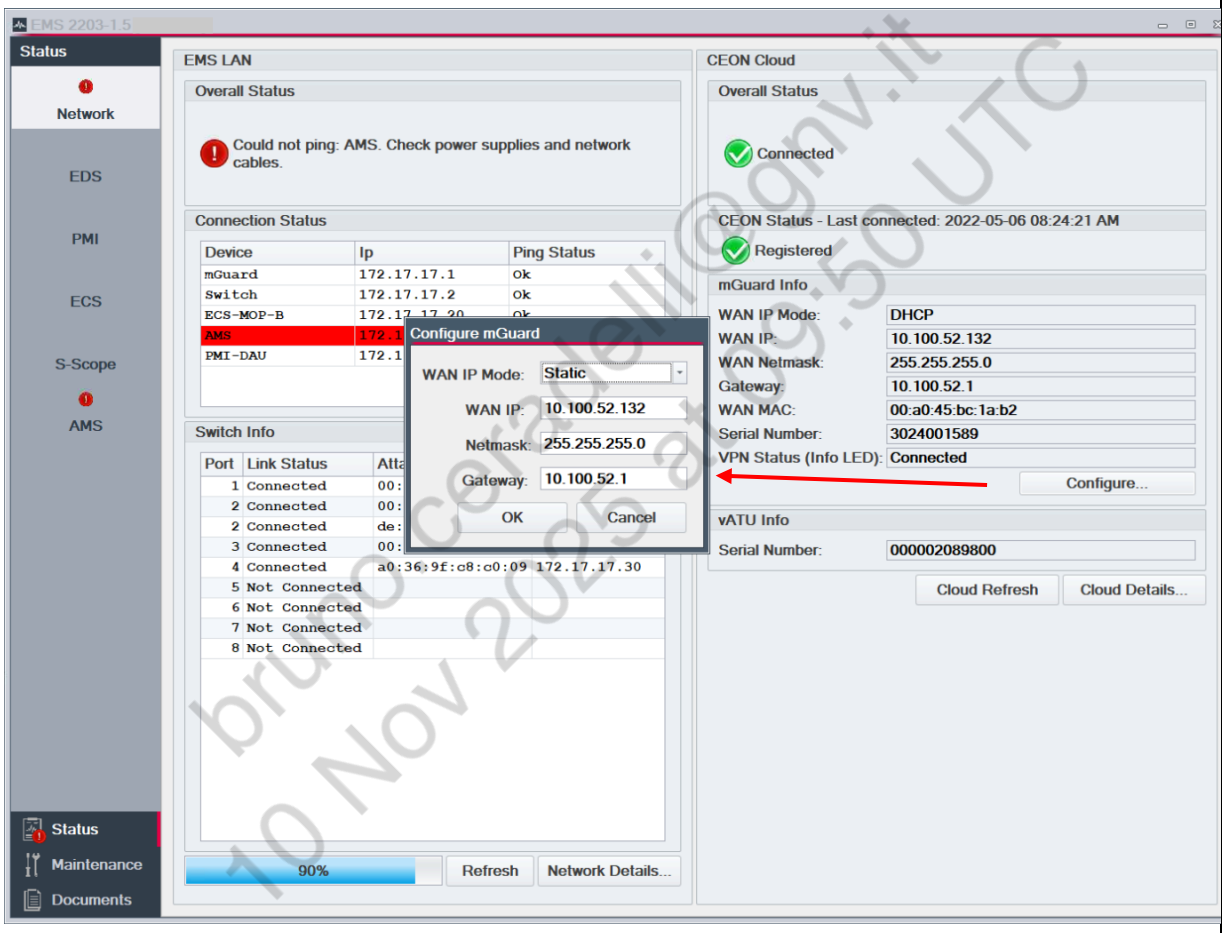
6.1- IP Settings (mGuard)

Scenario 1: WAN IP = DHCP

If WAN IP Mode is DHCP, the network will automatically assign an IP address, which should be informed to vessel's IT department for proper configuration of the network rules.

Scenario 2: WAN IP = Static

A static IP can only be determined by owner/vessel IT (MAN ES cannot inform you of the IP address).
Press **Configure** and type in the assigned address as shown in the example below.
You cannot leave the fields WAN Netmask and Gateway blank.



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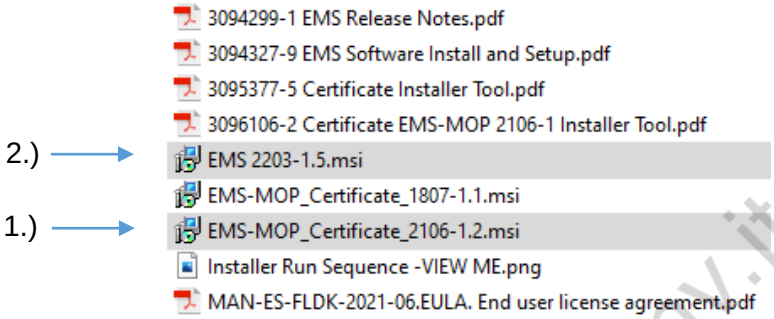
Please allow 30 seconds after pressing OK for the address to change.

EMS software update

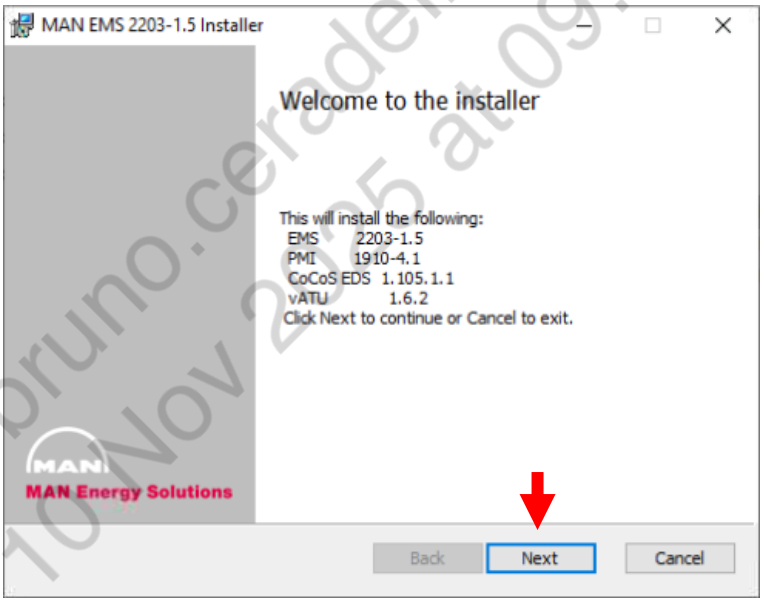
If the EMS software needs to be updated, MAN ES will provide a download link to a ZIP-file containing all necessary files. Extract all the files before opening the installation files.

1.) Run the "EMS-MOP Certificate" file as shown below and click **Next.** Then **Finish.**



2.) Run the EMS installer .msi file.

Allow 10-20 seconds for the installer window to appear and press **Next**.



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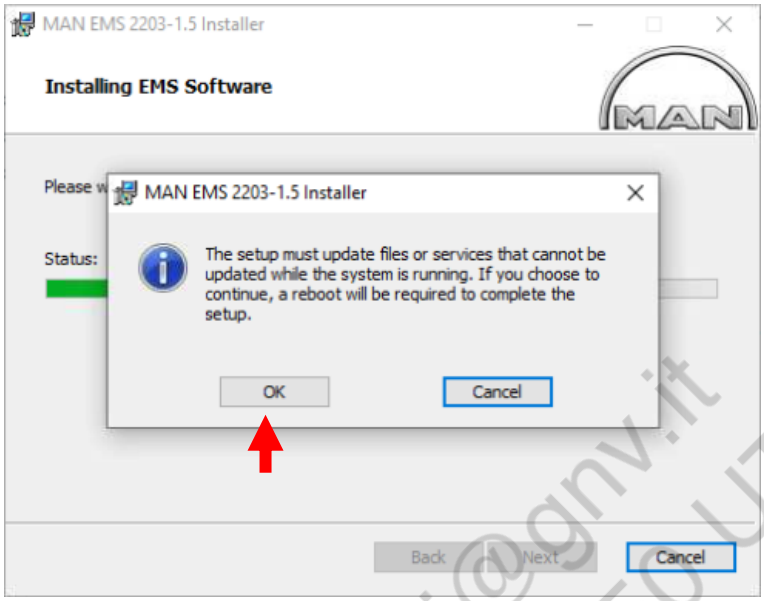
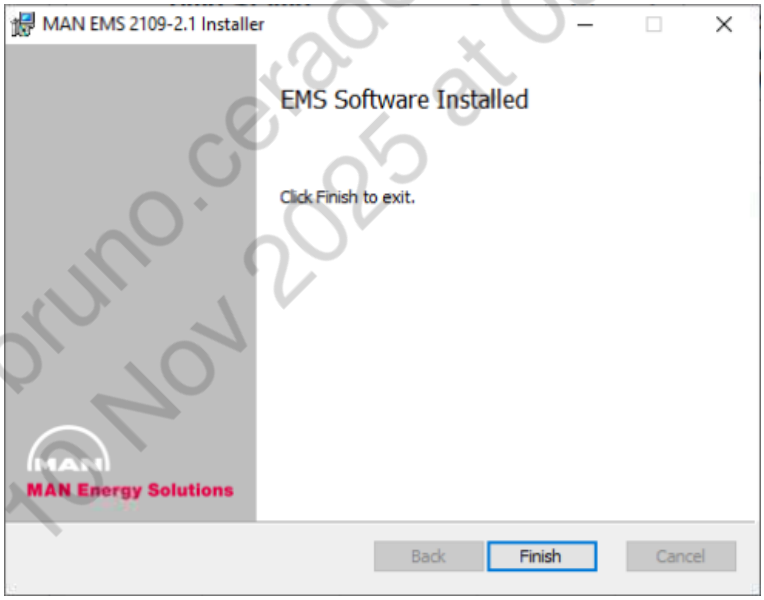
The following dialogue box will appear. Press **OK** and the EMS will restart automatically.



Once the installation is complete, press **Finish.**



PMI Auto-tuning, CoCos-EDS and EMS application will automatically open.
The installation is now complete.

This update procedure applies to all versions!

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Network requirements for MAN CEON

Please forward this checklist to your IT department/SAT provider.

1. A static or DHCP provided address that enables the mGuard to connect to the internet. (IP settings can be pre-configured or configured locally on the EMS).

2. A network cable (CAT6 or 6A) connection from mGuard WAN port to vessel switch (ECR switch).

3. Continuous connectivity (whitelisting) to communications gateway (ECR switch) installed on the ship's network needs to be provided to ALL of the following destination domains/IPs:

	Opens for	IP	Port number	Port type
3a	CEON access	35.156.77.222 35.157.233.80 35.156.61.78	443	TCP
3b	Firewall configuration and updates	149.247.176.129	443	TCP
3c	Time sync	91.189.94.4 91.189.91.157 35.156.77.222 35.157.233.80 35.156.61.78	123	UDP
3d	Remote access with IPSec VPN using NAT Traversal	149.247.177.99	500 4500	UDP

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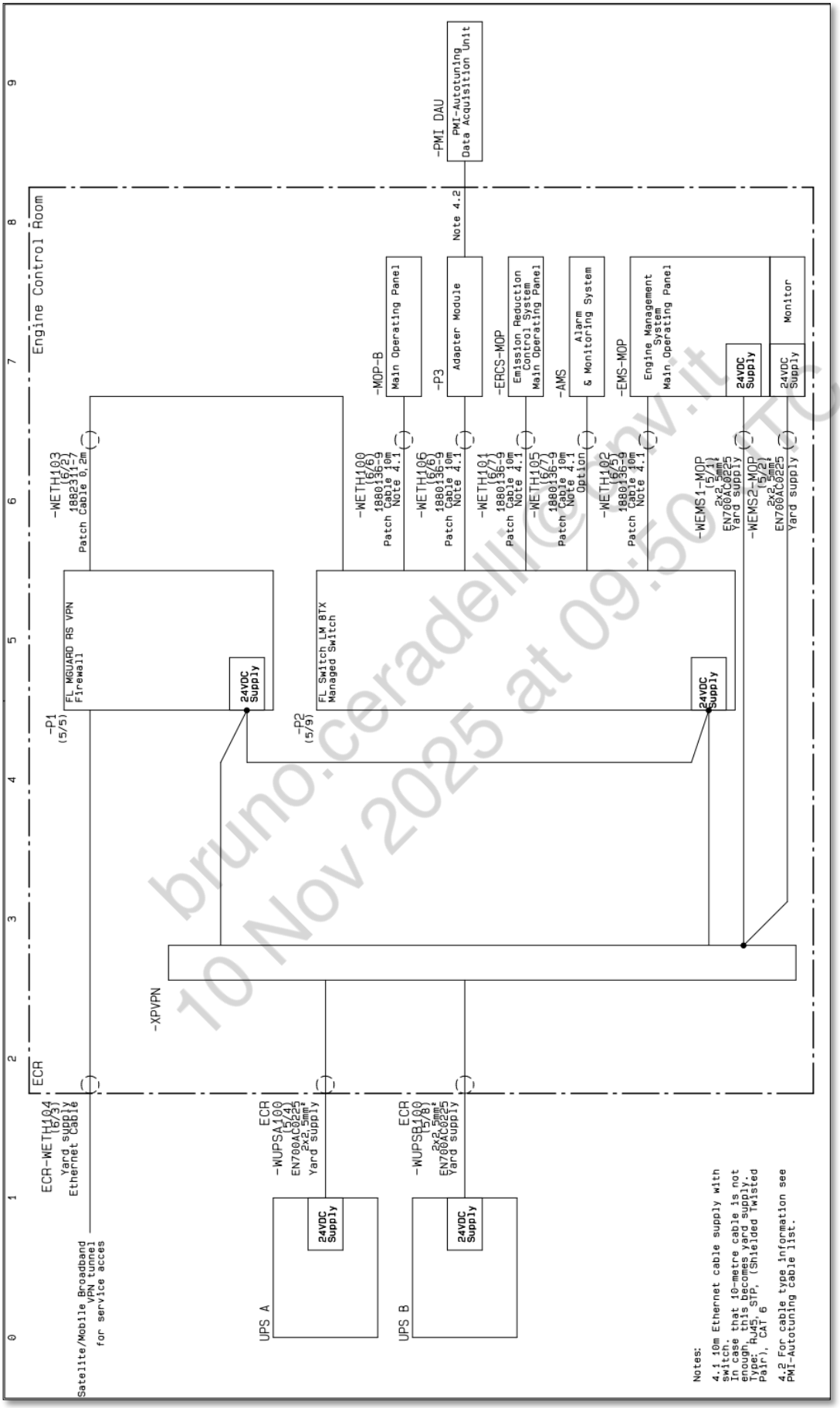
Appendix 1 – ECR UPS power supply



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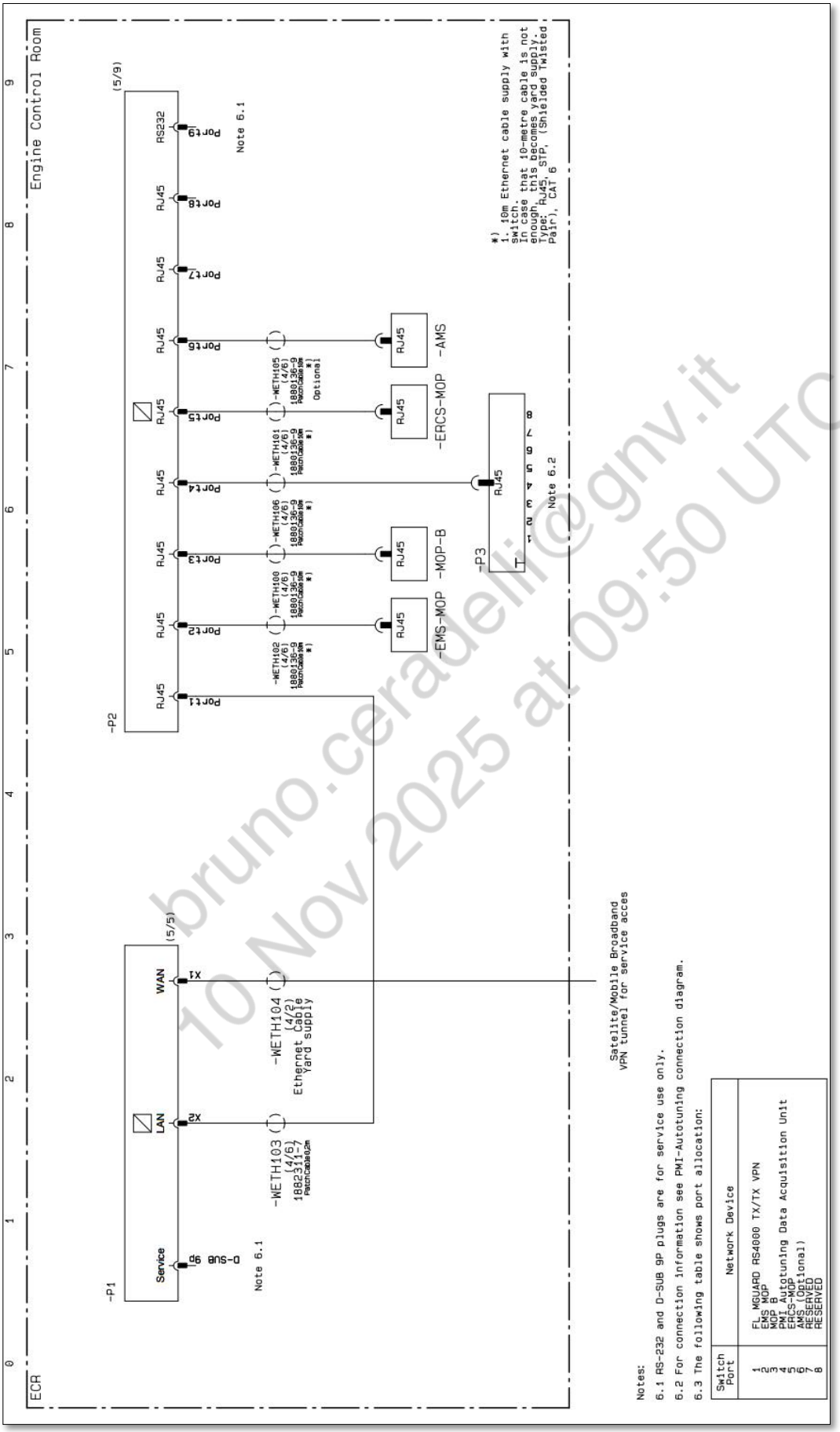
Appendix 2 – ECR Local Area Network



Notes:

6.1 RS-232 and D-SUB 9P plugs are for service use only.

6.2 For connection information see PMI-Autotuning connection diagram.

6.3 The following table shows port allocation:

Switch Port	Network Device
1	FL MGUARD RS4000 TX/TX VPN
2	EMS MOP
3	EOB
4	PMI Autotuning Data Acquisition Unit
5	ERCS-MOP
6	AMS (Optional)
7	RESERVED
8	RESERVED

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